

ASTRONOMY NOTES

Radius of earth: 6371 km

Circumference of earth: 40,075 km

Distance between Earth and Moon: 384,000 km

Sun's Circumference: 4,366,813 km

Earth's average distance b/w Earth and Sun: 150 million km

(Astronomical unit)

Distance:

Sun - Jupiter 5 AU

Sun - Saturn 9.58 AU

Sun - Neptune 30 AU

1 Light year: 9.5×10^{12} km / 63,241 AU

Andromeda galaxy from earth: 2.5 million ly

Superclusters size: 100 million to 700 million ly

10,000 to 100,000 galaxies in superclusters

Earth's speed around the sun: 30 km/sec

Moon speed around earth: 3683 km/h

Solar system's speed around galactic center: 230 km/s

Inner planets:

rocky planets are called Telluric planets

Outer planets:

gaseous planets are called Jovian planets

Hydrostatic Equilibrium:

State of balance where the inward pull of gravity is perfectly countered by an outward pushing force, preventing an object from collapsing or exploding.

Diffuse Emission:

Diffuse emission is defined as electromagnetic radiation that originates from an extended, spread out region of space rather than from a single, sharp point source like a star.

Gamma ray has a very large energy range.

Highest confirmed gamma-ray energy: 2.5×10^{15} eV (This is per photon)

Cascade Of Particles:

A cascade of particles is a rapid chain reaction where one high energy particle collides with matter and splits into many secondary particles, which then split further to create an avalanche of thousands or millions of particles.

One super fast, high-energy particle (like photon) crashes into a piece of matter (like an atom), Instead of just bouncing off, that heavy impact turns energy into matter. That energy turns into brand new pair of matter and antimatter.

These new particles are still moving fast enough to crash into other atoms. The energy turns to matter again. This happens over and over in a split second, turning one particle into a shower of millions.

A planet's rotation speed is related to how much matter it acquired when the Solar System formed → conservation of angular momentum

Sun doesn't obey the above rule because it emits photons and charged particles, Thus it loses angular momentum resulting in slow rotation rate.

Uranus's speed is slower than Neptune because it probably had collision with a massive object in past.

Point Source:

A point source is any celestial object that appears so incredibly small in the sky that it has no measurable size or shape. It is a single, concentrated pixel of light.

extended source:

An extended source is a celestial object that spans a measurable area of the sky, possessing a distinct shape, width, and surface structure.

An object is not permanently locked into one category. It depends completely on distance and power of the instrument observing it.

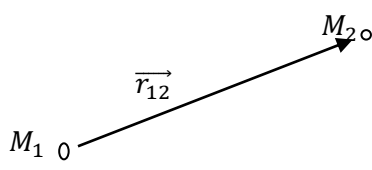
Unit Vector:

A unit vector is a special type of vector that has magnitude of exactly 1 and points in a specific direction. It is written with a hat (^) like $\hat{r}_{12}$

Because its length (magnitude) is exactly 1, a unit vector does not change the value or strength of any number its multiplied by. Its sole mathematical purpose is to specify direction in space

You can turn any standard vector into unit vector by taking full vector and dividing it by its own total length (magnitude) $\hat{r} = \vec{r} / |\vec{r}|$ Example: If a displacement vector $\vec{r}$ means "walk 5 meter East," its magnitude $|\vec{r}|$ is 5 meters. If you divide the vector by 5, you get a unit vector $\hat{r}$ that simply represents direction with a value of 1.

Newton's Law Of Gravitation:



The diagram shows two points representing masses, M_1 and M_2 . A vector labeled $\vec{r}_{12}$ points from M_1 to M_2 .

To calculate force on M_2 due to M_1 :

$$\vec{F}_{12} = \frac{-GM_1M_2 \hat{r}_{12}}{r_{12}^2} \quad G = 6.67 \times 10^{-11}$$

Annotations for the equation:

- $\vec{F}_{12}$: Tells force on M_2 by M_1
- $\hat{r}_{12}$: Tells direction
- r_{12}^2 : distance between masses
- G : Gravitational constant

Point Mass:

A point mass is an idealized object in physics that has a specific mass but occupies absolutely zero space. It is treated as a single, infinitely small mathematical point with no height, width, or depth.

Gravitational Field

The gravitational field due to mass M is the gravitational force per unit mass experienced by another object.

$$\vec{g} = \frac{\vec{F}}{m} = -\frac{GM\hat{r}}{r^2}$$

Force vs Gravitational Field Strength:

To define force, it's important to have 2 masses. To define gravitational field strength, we need one mass. Field is just a property of that mass. It tells us if a second object is at a distance x, how much gravitational force will the object experience.

For just magnitude:

$$g = \frac{GM}{r^2}$$

For a point on the surface of the earth:

$$\frac{(6.674 * 10^{-11}) * (\text{mass of earth})}{(\text{radius of earth})^2}$$

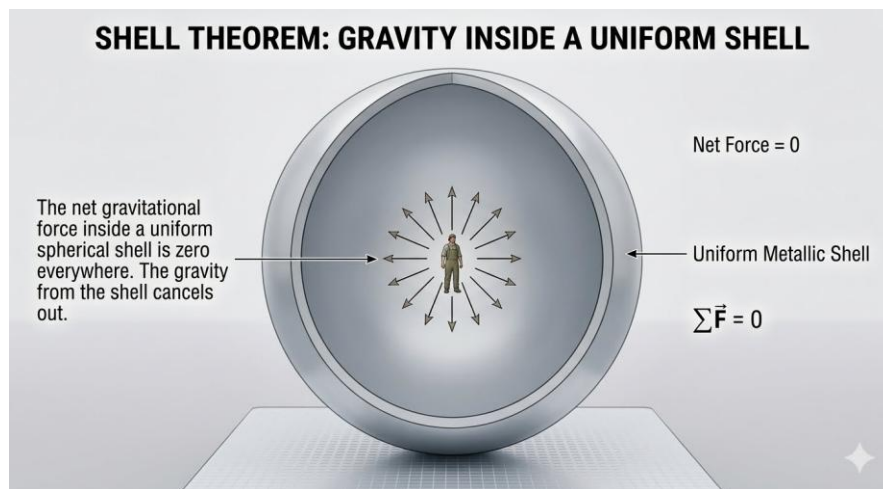
Gravitational field strength range is infinite though it gets very weak almost negligible over long distances.

Shell Theorem:

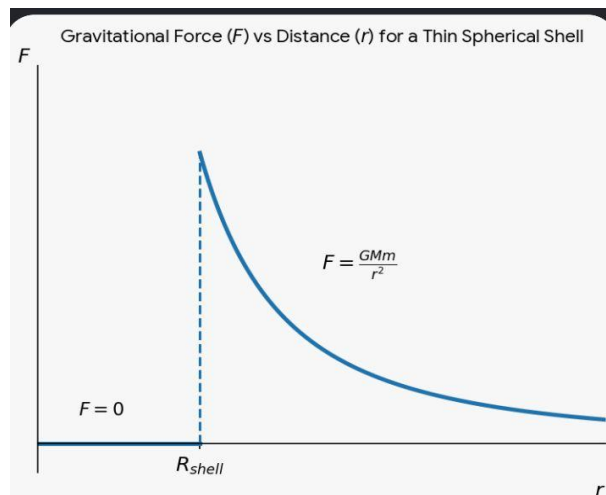
In a spherical symmetric shell

- 1) Inside a spherically symmetric shell, no resultant gravitational force will be exerted on an object
- 2) Outside the shell, it will exert a force on a body as if its entire mass were concentrated at the center.

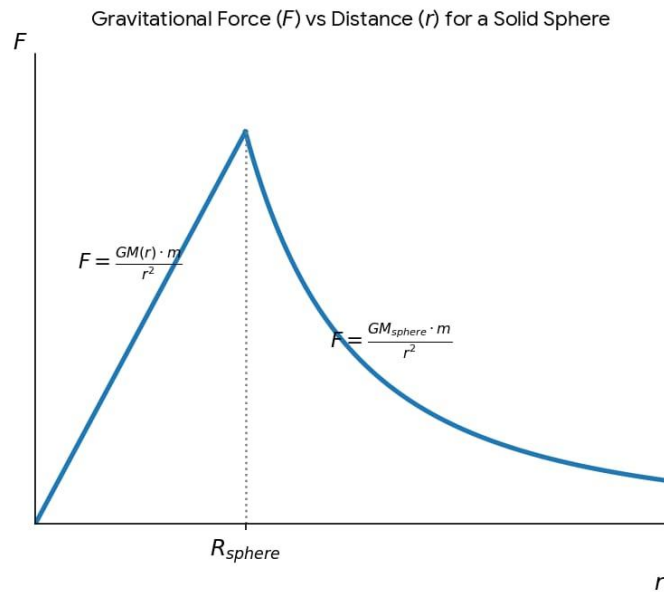
An object inside a spherical shell will experience no resultant force because gravitational forces from every direction cancel each other out.



For hollow sphere or shell:



For a solid sphere the force on an object inside will depend on r



For inside a solid sphere:

$$F = \frac{GM(r)m}{r^2}$$

If you are at the surface of Earth, the close mass $M(r)$ is the total mass of earth.

$$M(r) = \int_0^r \rho(r) 4\pi r^2 dr$$

Cumulative mass inside r
density at r

Special case if density is constant:

$$M(r) = \frac{4\pi r^3 \rho}{3}$$

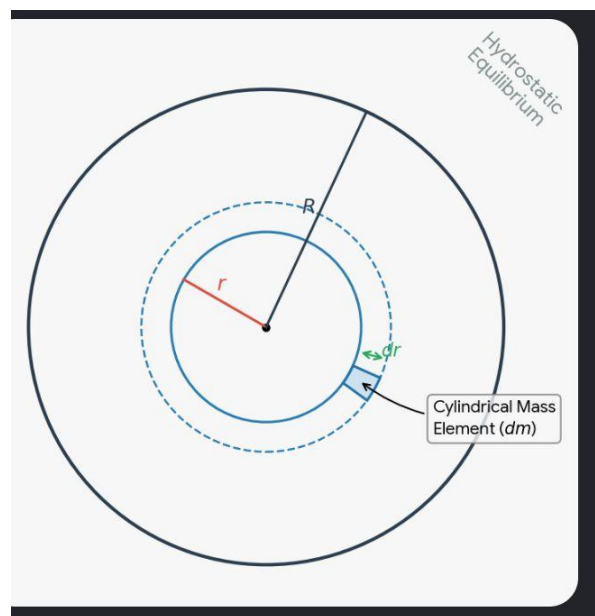
$$F = \frac{G M m r \pi \rho}{3}$$

$$F = \frac{G M m}{r^2}$$

$$F = \frac{G m \frac{4\pi r^3 \rho}{3}}{r^2}$$

$$F = \frac{4 G m \pi r \rho}{3}$$

Hydrostatic Equilibrium:



The mass of cylinder is dm

The thickness of shell is dr

The surface area of cylinder is dA

Volume of cylinder = dv

$$dv = dA \cdot dr$$

$$dm = \rho(r) dA dr$$

Force counter balancing gravitational force:

$$P = F/A$$

$$\begin{aligned}\text{Force due to pressure on top of cylinder} &= \vec{F}_{pt} \\ &= -P(r + dr) dA \hat{r}\end{aligned}$$

$$\begin{aligned}\text{Force due to pressure on the bottom of cylinder} &= \vec{F}_{pb} \\ &= P(r) dA \hat{r}\end{aligned}$$

Total force on dm due to pressure

$$\begin{aligned}\vec{F}_p &= P(r) dA \hat{r} - P(r + dr) dA \hat{r} \\ &= (P(r) - P(r + dr)) dA \hat{r} \\ &= -dP dA \hat{r}\end{aligned}$$

Total force on dm:

$$\begin{aligned}\vec{F} &= \vec{F}_g + \vec{F}_p \\ &= -\frac{GM(r) dm \hat{r}}{r^2} - dP dA \hat{r} \\ &= -\frac{GM(r) \rho(r) dA dr \hat{r}}{r^2} - dP dA \hat{r} \\ &= -\frac{GM(r) \rho(r) dA dr}{r^2} - dP dA\end{aligned}$$

$$F = ma$$

$$= dm \frac{d^2 r}{dt^2} \quad \text{or } dP / dt$$

$$= \rho(r) dA dr \frac{d^2 r}{dt^2}$$

$$\rho(r) dA dr \frac{d^2 r}{dt^2} = - \frac{GM(r) \rho(r) dA dr}{r^2} - dP dA$$

$$\rho(r) \frac{d^2 r}{dt^2} = - \frac{GM(r) \rho(r)}{r^2} - \frac{dP}{dr}$$

$$\frac{d^2 r}{dt^2} = 0$$

$$\frac{dp}{dr} = - \frac{GM(r) \rho(r)}{r^2}$$

Pressure is a decreasing function of r

Since $dP/dr < 0$, the pressure gradient balances the gravitational force

If gravitational pull increases, the star collapses or contracts.

If internal pressure increases, the star expands

$$\frac{dP}{dr} = - \frac{GM(r) \rho(r)}{r^2}$$

$$dP(r) = - \frac{GM(r) \rho(r) dr}{r^2}$$

$$\int_{P(r_1)}^{P(r_2)} dP(r) = \int_{r_1}^{r_2} - \frac{GM(r) \rho(r)}{r^2} dr$$

If density constant (unrealistic):

$$M(r) = 4\pi\rho \int_0^r r'^2 dr$$

$$= 4\pi\rho \frac{r^3}{3}$$

$$\begin{aligned}
P(r_2) - P(r_1) &= -G \int_{r_1}^{r_2} \left(\frac{4\pi\rho r^3}{3} \right) \frac{1}{r^2} \rho dr \\
&= - \frac{4G\pi\rho^2}{3} \int_{r_1}^{r_2} r dr \\
&= - \frac{4G\pi\rho^2}{3} \left[\left(\frac{r^2}{2} \right) \right]_{r_1}^{r_2} \\
&= - \frac{4G\pi\rho^2}{3} \left(\frac{r_2^2}{2} - \frac{r_1^2}{2} \right)
\end{aligned}$$

Equation of mass conservation:

$$\begin{aligned}
dM(r) &= 4\pi r^2 dr * \rho(r) \\
\frac{dM(r)}{dr} &= 4\pi r^2 \rho(r)
\end{aligned}$$

Luminosity: Energy emitted by a star per unit time

Luminosity symbol=L

Unit: J/s or watts

$$L_{\odot} = 3.828 * 10^{26} \text{ Watts}$$

In astro physics we do not use watts instead we use solar luminosity

For mass of sun in physics its written $M_{\odot}$

For stars on the main sequence:

- 1) Thermal gas pressure
- 2) Radiation pressure

For white dwarves and neutron star:

- 1) degeneracy pressure

Electron degeneracy (white dwarves)

Neutron degeneracy (neutron star)

Rough age of star:

$$\frac{M}{L} * 10^{10}$$

Mass is in $M_{\odot}$

Luminosity is in $L_{\odot}$

Four forces:

- 1) Strong Force
 - 2) Electromagnetic force
 - 3) Weak Force
 - 4) Gravitational
-

Equation of energy conservation:

$L(r)$ = luminosity of one layer at radius

$L(r)$ = net energy per unit time emitted by the layer

= energy emitted from all layer inside $L(r-dr)$ + energy emitted by layer at r – energy from inside absorbed in layer at $dL(r)$

$$\begin{aligned} dL(r) &= \varepsilon(r) dM(r) \\ &\nearrow = \varepsilon(r) 4\pi r^2 \rho(r) dr \end{aligned}$$

Net energy emitted per unit time per unit mass

$$\frac{dL(r)}{dr} = \varepsilon(r) 4\pi r^2 \rho(r)$$

Pressure = Momentum flux

$$dp = F dt$$

$$\int_{p_i}^{p_f} dp = \int_{t_i}^{t_f} F dt$$

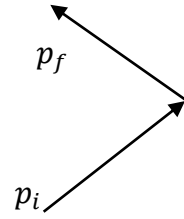


Only allowed if F is constant

$$p_f - p_i = F(t_f - t_i)$$

$$\Delta p = F(\Delta t)$$

STEP1: 1 Particle ,1 Collision



$$\vec{F}_{p \rightarrow w} = -\vec{F}_{w \rightarrow p}$$

$$\Delta \vec{p} = \vec{p}_f - \vec{p}_i = F_{w \rightarrow p} (t_{f1c} - t_{i1c})$$

$$\vec{p}_i = p_x \hat{i} + p_y \hat{j} + p_z \hat{k}$$

$$\vec{p}_f = p_x \hat{i} + p_y \hat{j} + p_z \hat{k}$$

$$P_x = p_{xf} - p_{xi} = -p_x - p_x = -2p_x$$

$$\Delta P_y = 0$$

$$\Delta p_z = 0$$

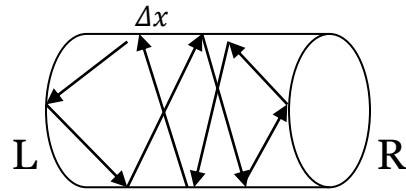
$$p_x \hat{i} + p_y \hat{j} + p_z \hat{k} = \vec{F}_{w \rightarrow p} \Delta t_1 c$$

$$-2p_x \hat{i} = \vec{F}_{w \rightarrow p} \Delta t_1 c$$

$$-2p_x \hat{i} = -F_{1c} \Delta t_1 c$$

$$F_{1c} = \frac{2p_x \hat{i}}{\Delta t_1 c}$$

Step2: 1 particle , 2 collisions



Particle bouncing again and again until it reaches R or L

$$\Delta t = t_f - t_1 > t_{f1c}$$

$$\vec{v}_i = v_{x_i}^A + v_{y_j}^A + v_{z_k}^A$$

$$\vec{v}_f = -v_{x_i}^A + v_{y_j}^A + v_{z_k}^A$$

Time taken by particle to go from R to L:

$$t_{RL} = \frac{\Delta x}{v_x}$$

$$t_{LR} = t_{RL}$$

$$\Delta t = t_{RL} + t_{LR}$$

$$= \frac{2\Delta x}{v_x}$$

$$F = \frac{2p_x \hat{i}}{\Delta t}$$

$$\vec{F} = \frac{\frac{2p_x \hat{i}}{\frac{2\Delta x}{v_x}}}{v_x}$$

$$\vec{F} = \frac{p_x v_{x_i}^A}{\Delta x}$$

Step3: Particle at speed v (averaged over time ,averaged over many particles)

$$\vec{v}_1 = v_{1x_i}^A + v_{1y_j}^A + v_{1z_k}^A$$

$$\overrightarrow{v_2}=v_{2x_i}^{\Lambda}+v_{2y_j}^{\Lambda}+v_{2zk}^{\Lambda}$$

$$\cdot$$

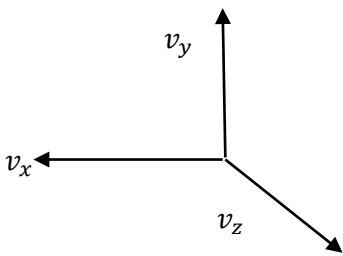
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$$\overrightarrow{v_N}=v_{Nx_i}^{\Lambda}+v_{Ny_j}^{\Lambda}+v_{Nzk}^{\Lambda}$$

$$\langle v_x^2\rangle=\langle v_y^2\rangle=\langle v_z^2\rangle$$

$$|\vec{v}|=\sqrt{v_x^2+v_y^2+v_z^2}$$



$$v^2=\langle v_x^2\rangle+\langle v_y^2\rangle+\langle v_z^2\rangle$$

$$=\text{ b}+\text{ b}+\text{ b}$$

$$=3\text{b}$$

$$v^2=3\langle v_x^2\rangle$$

$$\frac{v^2}{3}=\langle v_x^2\rangle$$

$$\frac{1}{N_v}\big[F,+F_2+\cdots+F_{N_v}\big]=\frac{m}{\Delta x}\frac{1}{N}\mathcal{V}^{[v,x^2+v_2x^2+v_Nx^2]}$$

$$\langle F(v)\rangle=\frac{m}{\Delta x}\langle v_x^2\rangle$$

$$\langle F(v)\rangle=\frac{mv^2}{3\Delta x}$$

$$\langle F(v)\rangle=\frac{pv}{3\Delta x}$$

Number of particles with momenta between $[p, p+dp]$:

Let $dN = N(p)dp$

Then the total number of particles of all momenta:

$$N = \int_0^{\infty} N(p) dp$$

Force exerted by particles with momenta in range $(p, p+dp)$:

$$dF = \langle F(P) \rangle dN$$

$$= \frac{1}{3} \frac{N(p) p v dp}{\Delta x A}$$

Step 4: for all particles

$$F = \int_0^{\infty} \frac{1}{3} \frac{N(p) p v dp}{\Delta x A}$$

$$P = F/A$$

$$F = \int_0^{\infty} \frac{1}{3} \frac{p v}{\Delta x A} N(p) dp$$

No of particles with momentum $(p, p+dp)$

Volume of cylinder

$$= \int_0^{\infty} \frac{1}{3} \frac{N(p)}{V} p v dp$$

$$n(p) dp = \frac{N(p) dp}{V}$$

number of particles per unit volume in range $(p, p+dp)$

$$P = \int_0^{\infty} \frac{1}{3} p v n(p) dp$$

Pressure Integral

Pressure ideal gas (non relativistic):

$$P = mv$$

$$P = \int_0^{\infty} \frac{1}{3} m v^2 n(v) dv$$

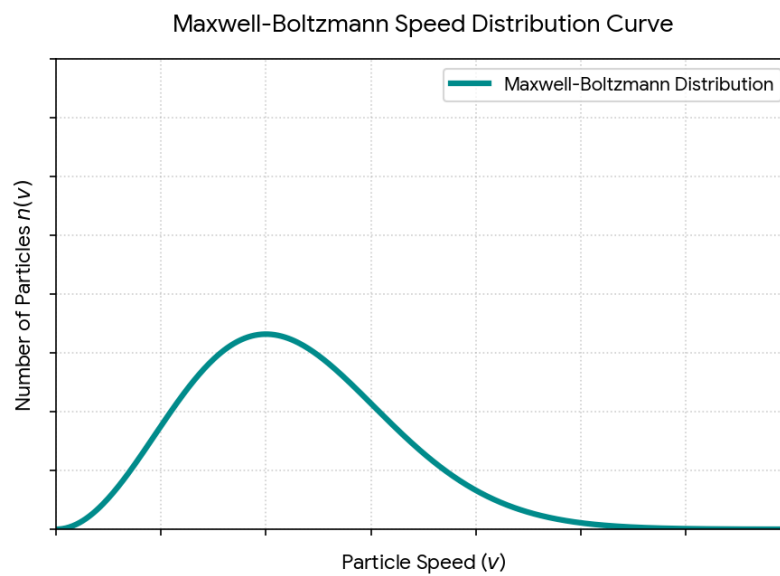
Maxwell Boltzman distribution

Number of particles per unit volume = n_v

T= temperature

$$\text{Boltzmann constant} = k_B = 8.62 * 10^{-5} \frac{eV}{K} = 1.38 * 10^{-23} \frac{J}{K}$$

$$n(v) = n_0 \frac{4}{\pi} \left(\frac{m}{2k_B T} \right)^{\frac{3}{2}} e^{-\left(\frac{mv^2}{2k_B T} \right)} v^2$$



$$P = \int_0^\infty \underbrace{\frac{n_0}{3} \frac{4m}{\sqrt{\pi}} \left(\frac{m}{2k_B T} \right)^{\frac{3}{2}}}_{\text{Constant}} e^{-\left(\frac{mv^2}{2k_B T} \right)} v^4 dv$$

Constant

$$= \int_0^\infty x^a e^{-bx^2} dx$$

$$x^a = v^4$$

$$b = \frac{m}{2k_B T}$$

$$a=4$$

$$= \int_0^\infty x^4 e^{-bx^2} dx$$

$$= \frac{3\sqrt{\pi}}{2^3 b^{\frac{5}{2}}}$$

$$= \frac{n_0}{3} \frac{4m}{\sqrt{\pi}} \left(\frac{m}{2k_B T} \right)^{\frac{3}{2}} \times \frac{3\sqrt{\pi}}{2^3 \left(\frac{m}{2k_B T} \right)^{\frac{5}{2}}}$$

$$= \frac{n_0}{3} \frac{m}{2} \frac{2k_B T}{m} x^3$$

$$= n_0 k_B T$$

$$P = n_0 k_B T$$

number of particles per unit volume

$$n_0 = N/V$$

Total Number of particles

volume

$$P = \frac{N}{V} k_B T$$

$$PV = N k_B T$$

$$n_0 = \frac{\rho}{\bar{m}} \leftarrow \text{average mass of particles}$$

Mean molecular weight $\longrightarrow \mu = \frac{\bar{m}}{M_H}$

mass of proton = $1.6375 \times 10^{-27} \text{ kg}$

mass of hydrogen atom (proton) $\longleftarrow$

$$P = \frac{\rho}{\bar{m}} k_B T$$

$$P = \frac{\rho}{\mu M_H} k_B T$$

Current composition sun's core:

H = 34 percent He = 64 percent heavier = 2 percent

Young star composition:

H = 74 percent He = 24 percent heavier elements = 2 percent

X, Y, Z are mass fractions

$$X = \frac{\text{Mass of H}}{\text{Total mass}}$$

$$Y = \frac{\text{Mass of He}}{\text{Total mass}}$$

$$Z = \frac{\text{Mass of heavier elements}}{\text{Total mass}}$$
